



## Analysis

## How does environmental performance affect financial performance? Evidence from Japanese manufacturing firms

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## ABSTRACT

This paper examines the effects of environmental performance on financial performance using the data of Japanese manufacturing firms from 2004 to 2008. As the environmental performance, our study considers the two different environmental issues of waste and greenhouse gas emissions in capturing the effects of corporate environmental management on financial performance. In addition, to clarify how each financial performance responds to a firm's effort in dealing with different environmental issues, we utilize many financial performance indices reflecting various market evaluations. Our estimation results show the different effects of each environmental performance on financial performance. Waste emissions do not generally have significant effects on financial performance. On the other hand, greenhouse gas reduction leads to an increase in financial performance in the whole sample and clean industries, although it does not have significant effects on financial performance in dirty industries. Furthermore, as the firm growth rate increases, the partial effects of waste emissions on financial performance decrease, whereas the partial effects of greenhouse gas emissions on financial performance increase.

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## 1. Introduction

Does better environmental performance improve a firm's financial performance? In seeking to answer this question, many studies have been conducted from both the economic and business administration perspectives (see e.g. Hart and Ahuja, 1996; King and Lenox, 2002; Konar and Cohen, 2001; Nakao et al., 2007; Porter and van der Linde, 1995; Russo and Fouts, 1997).<sup>1</sup> Researches on the relationship between environmental and financial performance are not only meaningful in the sense of analyzing firm behavior, but also important from the social benefit perspective. In economic literature, environmental problems have traditionally been treated as inconsistencies between social and private benefits and have mainly been left to government intervention to solve them. However, if financial performance is positively related to environmental performance, firms have incentives to reduce their environmental damages. This means that environmental problems may be solved by the market mechanism without government intervention, leading to a preferable environment for both firms and the government. For this reason,

analyzing the relationship between environmental and financial performance also has important policy implications.

In recent years, many stakeholders of firms such as governments, non-governmental organizations, local communities, consumers, trading partners, employees, investors, financial agencies, and stockholders are conscious of corporate environmental management, especially in developed countries. This directly or indirectly influences the financial performance of firms. For example, if a firm violates an environmental regulation or causes an environmental accident, the firm not only has to pay fines and penalties, but may suffer from a loss of trust and reputation or a boycott of goods. Such risks have negative effects on the evaluation of a firm's future profits. On the other hand, a firm that actively addresses environmental issues might gain positive reputation among some stakeholders and may influence them to expect that the firm will succeed in reducing environmental risks and production costs in the long term. Therefore, firms have an incentive to address various environmental issues against the backdrop of various stakeholders' interests.

This paper demonstrates the relationship between environmental and financial performance, and moreover, considers the relationship between the characteristics of each environmental issue and the responses of various markets and stakeholders' behaviors behind them. In previous empirical studies, the relationships between environmental and financial performance are controversial. Cordeiro and Sarkis (1997), Sarkis and Cordeiro (2001), Wagner et al. (2002),

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E-mail address: [keisuke.okada.1125@gmail.com](mailto:keisuke.okada.1125@gmail.com) (K. Okada).<sup>1</sup> Murphy (2002), Molina-Azorin et al. (2009), and Blanco et al. (2009) provide detailed surveys on the relationship between environmental and financial performance.

and Rassier and Earnhart (2010) advocate a partially or completely negative relationship.<sup>2</sup> On the other hand, Hart and Ahuja (1996), Russo and Fouts (1997), Konar and Cohen (2001), and King and Lenox (2002) report a positive relationship. Some studies such as Hart (1995) and Ambec and Lanoie (2008) indicate the channels from which environmental performance affects financial performance. Specifically, Hart (1995) explains the relationship between the two performance by considering natural resources in the resource-based view. Ambec and Lanoie (2008) summarize the channels and suggest that better environmental performance can improve financial performance in both revenue (e.g. better access to certain markets; differentiating products; and selling pollution-control technologies) and cost aspects (e.g. risk management and relations with external stakeholders; costs of material, energy, and service; cost of capital; and cost of labor).

Meanwhile, most of the existing studies do not necessarily capture the characteristics of each environmental issue enough since they use a few indices as the proxy for environmental performance. In recent years, there have been many kinds of environmental issues such as global warming, acid rain, deforestation, ozone depletion, biodiversity, pollution of the environment by toxic chemical compounds, and waste issues. Each environmental issue has different characteristics such as the scope of pollution (e.g. local or global), length of time until damages emerge, severity of the damages, facilities for specifying the polluters, and existence of regulations and international treaties. These various characteristics suggest that different stakeholders may place emphasis on different environmental issues. Some stakeholders, for example a local community, may directly suffer from a firm's environmental pollution while others may not suffer from it but may have monetary relationship with the firm. Therefore, some stakeholders may think that global warming is a more important problem than any other environmental issue and others may think that the waste problem is the most crucial issue. These stakeholders' differentiated preferences for environmental issues may affect financial performance. Taking this into account, this paper therefore considers two different environmental issues, waste and greenhouse gas emissions, as environmental performance. Waste and greenhouse gas emissions are different in many points such as the scope of pollution and existence of regulations, and therefore they are related to different kinds of stakeholders who directly or indirectly influence the financial performance of firms through various markets. Employing the amount of waste and greenhouse gas emissions enables us to investigate the difference of the market evaluation on different kinds of environmental issues.

Previous studies using the amount of emissions as environmental performance are as follows. Hart and Ahuja (1996) show that a reduction in emissions of selected pollutants improves financial performance, such as return on sales (ROS), return on assets (ROA), and return on equity (ROE) for a sample of S&P 500 firms. Konar and Cohen (2001) also indicate that a decrease in toxic chemical emissions ameliorates Tobin's  $q - 1$  in the S&P 500 firms.<sup>3</sup> King and Lenox (2002) obtain the same conclusion using toxic chemicals emissions as environmental performance, and ROA and Tobin's  $q$  as financial performance for a sample of publicly traded U.S. manufacturing firms. In contrast, Wagner et al. (2002) make an index for environmental performance based on SO<sub>2</sub>, NO<sub>x</sub>, and Chemical Oxygen Demand (COD) emissions, and use ROS, ROE, and return on capital employed (ROCE) as financial performance for a sample of firms in the European paper manufacturing industry. They show that worsened environmental

performance improves ROCE and has no significant effects on ROS and ROE.<sup>4</sup>

As for financial performance, this paper pays attention to many indices such as ROE, ROA, return on investment (ROI), return on invested capital (ROIC), ROS, Tobin's  $q - 1$ , and the natural logarithm of Tobin's  $q$  in order to take into account each market's evaluation of corporate management dealing with different environmental issues. In existing research studies, financial performance is proxied by some indices such as ROA, ROE, and ROS by Hart and Ahuja (1996); ROA by Russo and Fouts (1997); Tobin's  $q - 1$  and the natural logarithm of Tobin's  $q$  by Konar and Cohen (2001); ROS by Sarkis and Cordeiro (2001); ROA and Tobin's  $q$  by King and Lenox (2002); ROCE, ROE, and ROS by Wagner et al. (2002); ROA, ROS, and Tobin's  $q$  by Elsayed and Paton (2005); and ROA and Tobin's  $q - 1$  by Nakao et al. (2007). These studies provide mixed results on the relationship between environmental and financial performance, implying that there are various channels from which environmental performance affects financial performance. Given this, the important point is that there is a possibility that each financial performance is affected directly or indirectly by different stakeholders.

This paper focuses on the Japanese manufacturing sector using the data of 268 firms from 2004 to 2008. Since the manufacturing sector is generally considered to have a closer relationship with environmental issues, we consider that manufacturing firms have stronger incentives to actively or voluntarily deal with environmental issues through various means. In practice, active environmental management such as over-compliance with regulations, publication of Corporate Social Responsibility (CSR) reports, and other voluntary approaches are introduced in many Japanese firms. With regard to previous studies using Japanese firm-level data, Nakao et al. (2007) use the environmental score from the Nikkei Environmental Management Survey as environmental performance and show that better environmental performance improves financial performance such as ROA and Tobin's  $q - 1$  in the Japanese manufacturing sector. Nagayama and Takeda (2007), Yamaguchi (2008), and Takeda and Tomozawa (2008) also investigate the effects of new environmental information (e.g. release of environmental management ranking) on stock prices by using an event study methodology. Furthermore, Cole et al. (2006) analyze the determinants of environmental performance of firms in Japan and find that globalization has positive impacts on environmental management.

Finally, our contributions to literature are as follows. First, in order to capture the characteristics of different environmental issues, we pay attention to both waste and greenhouse gas emissions as environmental performance. Second, to clarify how various financial performance is influenced by different environmental issues, we use seven financial performance indices such as ROE, ROA, ROI, ROIC, ROS, Tobin's  $q - 1$ , and the natural logarithm of Tobin's  $q$ , and examine the behaviors of various stakeholders. Third, to the best of our knowledge, this paper is the first in attempting to utilize the amount of emissions as environmental performance instead of environmental scores from studies using Japanese firm-level data.

The remainder of this paper is organized as follows. Section 2 discusses the estimation methodology and data. Section 3 presents empirical results. Section 4 concludes the paper.

## 2. Estimation Methodology and Data

The main purpose of this study is to examine the effects of environmental performance on financial performance. Following

<sup>2</sup> Walley and Whitehead (1994) theoretically explain a negative relationship between environmental and financial performance.

<sup>3</sup> Konar and Cohen (2001) also use the number of lawsuits as environmental performance.

<sup>4</sup> As the proxy for environmental performance, some previous studies use integrated indices provided by independent organizations such as rating companies, instead of the amount of emissions. For example, Russo and Fouts (1997), Butz and Plattner (2000), Salama (2005), and Elsayed and Paton (2005) utilize environmental ranking as environmental performance.

**Table 1**  
Environmental performance and financial performance (full sample).

|                   | (1)                    | (2)                    | (3)                    | (4)                    | (5)                    | (6)                    | (7)                    |
|-------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                   | ROE                    | ROA                    | ROI                    | ROIC                   | ROS                    | $q - 1$                | $\ln(q)$               |
| Size              | 0.0046<br>(0.0029)     | 0.0030**<br>(0.0012)   | 0.0020<br>(0.0019)     | 0.0047**<br>(0.0023)   | 0.0017<br>(0.0039)     | 0.1507***<br>(0.0153)  | 0.1063***<br>(0.0097)  |
| Growth            | 0.0001<br>(0.0003)     | 0.0002<br>(0.0001)     | −0.0001<br>(0.0001)    | −0.0002<br>(0.0001)    | 0.0001<br>(0.0004)     | 0.0003<br>(0.0023)     | 0.0004<br>(0.0015)     |
| R&D               | −0.2962***<br>(0.1054) | −0.1039**<br>(0.0406)  | −0.1595**<br>(0.0645)  | −0.2071***<br>(0.0713) | 0.0486<br>(0.4475)     | 0.7543<br>(0.6522)     | 0.4781<br>(0.4101)     |
| Advertisement     | −0.2243<br>(0.1772)    | 0.0665<br>(0.0777)     | 0.0178<br>(0.1071)     | 0.1998<br>(0.1463)     | −0.3225<br>(0.7811)    | 3.0918***<br>(1.0465)  | 2.5184***<br>(0.6113)  |
| Capital intensity | −0.0318***<br>(0.0103) | 0.0012<br>(0.0021)     | 0.0033<br>(0.0029)     | 0.0120**<br>(0.0051)   | −0.0315***<br>(0.0066) | −0.0171<br>(0.0272)    | −0.0157<br>(0.0149)    |
| Leverage          | 0.0026<br>(0.0114)     | −0.0127***<br>(0.0023) | −0.0238***<br>(0.0035) | −0.0276***<br>(0.0051) | 0.0140**<br>(0.0063)   | −0.0825***<br>(0.0319) | −0.0278<br>(0.0172)    |
| Waste             | −0.0033<br>(0.0024)    | 0.0009<br>(0.0011)     | −0.0004<br>(0.0019)    | −0.0005<br>(0.0023)    | 0.0235<br>(0.0206)     | 0.0161<br>(0.0163)     | 0.0198<br>(0.0125)     |
| Greenhouse gas    | −0.0147<br>(0.0205)    | −0.0139**<br>(0.0055)  | −0.0228***<br>(0.0064) | −0.0281***<br>(0.0083) | −0.0254<br>(0.0490)    | −0.0794*<br>(0.0458)   | −0.0410<br>(0.0401)    |
| Constant          | 0.0630*<br>(0.0324)    | 0.0489***<br>(0.0150)  | 0.1023***<br>(0.0242)  | 0.0848***<br>(0.0277)  | 0.0474*<br>(0.0262)    | −1.2638***<br>(0.1631) | −0.9718***<br>(0.1124) |
| R <sup>2</sup>    | 0.13                   | 0.14                   | 0.20                   | 0.12                   | 0.14                   | 0.18                   | 0.21                   |
| No. of firms      | 268                    | 268                    | 268                    | 268                    | 268                    | 268                    | 268                    |
| Observations      | 751                    | 751                    | 751                    | 751                    | 751                    | 749                    | 749                    |

Notes:

1. The numbers in parentheses are the heteroskedasticity-robust standard errors.
2. The asterisks \*\*\*, \*\*, and \* indicate 1%, 5%, and 10% of significance levels, respectively.

previous research studies, some other determinants that may have effects on financial performance are also included in the estimation equation as explanatory variables in addition to environmental performance. Our basic specification is expressed as follows:

$$\begin{aligned} \text{Financial performance}_{it} = & \beta_0 + \beta_1 \text{Size}_{it} + \beta_2 \text{Growth}_{it} + \beta_3 \text{R\&D}_{it} \\ & + \beta_4 \text{Advertisement}_{it} + \beta_5 \text{Capital intensity}_{it} \\ & + \beta_6 \text{Leverage}_{it} + \beta_7 \text{Waste}_{it} \\ & + \beta_8 \text{Greenhouse gas}_{it} + \mu_j + \varepsilon_{it}, \end{aligned}$$

where  $i$  denotes the firm;  $j$  shows the industry; and  $t$  indicates the period.  $\mu$  is the industry-specific fixed effects, as mentioned below, and  $\varepsilon$  is the standard error term. *Financial performance* is ROE, ROA, ROI, ROIC, ROS, Tobin's  $q - 1$ , or the natural logarithm of Tobin's  $q$ . We use these dependent variables because each of them reflects the behavior and evaluation of various stakeholders with different interests. ROE includes the stockholders' evaluation and performance of the goods market. ROA, ROI, and ROIC reflect not only the equity capital contributed by stockholders but also borrowed capital provided by creditors and investors. ROS indicates the market evaluation by consumers and trading partners. Tobin's  $q - 1$  and the natural logarithm of Tobin's  $q$  are interpreted as the intangible assets value of the firms.<sup>5</sup> Then, as for explanatory variables, *Size* represents the firm size; *Growth* is the firm growth; *R&D* is the research and development intensity; *Advertisement* is the advertisement intensity; *Capital intensity* is defined as the sales and operating revenue divided

by stockholders' equity; and *Leverage* is the financial leverage, defined as the sum of liabilities and net assets divided by total stockholders' equity. The detailed definitions of each variable are provided in Appendix Table A1. The rationale underlying our selection of these variables is based on previous preeminent studies such as Russo and Fouts (1997), Konar and Cohen (2001), King and Lenox (2002), and Nakao et al. (2007).

As the proxies for environmental performance, we take waste and greenhouse gas emissions into account. *Waste* and *Greenhouse gas* are defined as waste and greenhouse gas emissions divided by sales and operating revenue, respectively. We pay attention to these two factors because they reflect different kinds of environmental problems. In Japan, waste emissions are regulated by various laws. Many environmental pollution issues emerged during the high economic growth period of the 1950s through 1970s. Given that these issues aroused national discussion on environmental problems, various environmental laws and regulations were formulated such as the Basic Law for Environmental Pollution Control in 1967, which was succeeded by the Environment Basic Law in 1993, and the Basic Act on Establishing a Sound Material-Cycle Society in 2000. On the other hand, greenhouse gas emissions have only been paid attention to in relatively recent years since the address of global warming issues, which lead to the formulation of the Law Concerning the Promotion of the Measures to Cope with Global Warming in 1998. However, this law does not explicitly address the reduction in greenhouse gas emissions of individual firms. Since more regulations on greenhouse gas emissions may be imposed in the future due to this insufficiency, individual firms need to deal with this uncertainty. In particular, waste and greenhouse gas have different characteristics as environmental issues, such as the scope of pollution and existence of regulations.

Our study focuses especially on the sign and significance of each environmental performance on various financial performance, which is influenced by various stakeholders' behaviors. If environmentally-friendly management has positive (negative) effects on financial performance, the sign of environmental performance can be negative (positive). In addition, if environmentally-friendly management is not related to financial performance, the coefficient is not significant.

<sup>5</sup> Based on Konar and Cohen (2001), the concept of Tobin's  $q$  can be briefly explained as follows. The market value of the firm ( $MV$ ) can be expressed as the summation of the firm values from the tangible assets ( $V_T$ ) and from the intangible assets ( $V_I$ ). Since Tobin's  $q$  is defined as  $MV/V_T$ , from a simple calculation, Tobin's  $q - 1$  is equal to  $V_I/V_T$ . Therefore, Tobin's  $q - 1$  is interpreted as the intangible assets value of the firms, such as patents, brand name, and so on. Following Hirsch and Seaks (1993) and Konar and Cohen (2001), we use Tobin's  $q - 1$  and the natural logarithm of Tobin's  $q$  as the dependent variable. Hirsch and Seaks (1993) compare the two specifications with respect to Tobin's  $q$  using Box–Cox transformations and show that the semi-log specification has higher log-likelihood values than the linear specification.

**Table 2**  
Environmental performance and financial performance (nine clean industries).

|                   | (1)                    | (2)                    | (3)                    | (4)                    | (5)                    | (6)                    | (7)                    |
|-------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                   | ROE                    | ROA                    | ROI                    | ROIC                   | ROS                    | $q - 1$                | $\ln(q)$               |
| Size              | 0.0063*<br>(0.0038)    | 0.0030**<br>(0.0014)   | 0.0022<br>(0.0020)     | 0.0048*<br>(0.0026)    | 0.0035<br>(0.0051)     | 0.1346***<br>(0.0171)  | 0.0979***<br>(0.0111)  |
| Growth            | −0.0001<br>(0.0004)    | 0.0000<br>(0.0002)     | 0.0001<br>(0.0002)     | −0.0001<br>(0.0003)    | −0.0003<br>(0.0004)    | −0.0025<br>(0.0020)    | −0.0014<br>(0.0013)    |
| R&D               | −0.3461**<br>(0.1355)  | −0.1573***<br>(0.0581) | −0.2488***<br>(0.0893) | −0.3160***<br>(0.1021) | −0.1987<br>(0.6340)    | 0.3127<br>(0.8114)     | 0.1583<br>(0.5338)     |
| Advertisement     | −0.2490<br>(0.3081)    | 0.1920<br>(0.1178)     | 0.2252<br>(0.1393)     | 0.4649**<br>(0.2131)   | −0.7181<br>(1.3723)    | 4.0408**<br>(1.5700)   | 2.9974***<br>(0.9403)  |
| Capital intensity | −0.0335***<br>(0.0119) | 0.0009<br>(0.0024)     | 0.0017<br>(0.0032)     | 0.0107*<br>(0.0059)    | −0.0307***<br>(0.0076) | −0.0303<br>(0.0311)    | −0.0241<br>(0.0164)    |
| Leverage          | 0.0031<br>(0.0139)     | −0.0116***<br>(0.0026) | −0.0204***<br>(0.0039) | −0.0243***<br>(0.0060) | 0.0119*<br>(0.0068)    | −0.0557<br>(0.0360)    | −0.0121<br>(0.0190)    |
| Waste             | −0.0058*<br>(0.0032)   | 0.0011<br>(0.0013)     | 0.0000<br>(0.0023)     | 0.0003<br>(0.0026)     | 0.0089<br>(0.0175)     | −0.0037<br>(0.0125)    | 0.0064<br>(0.0072)     |
| Greenhouse gas    | −0.0171<br>(0.0225)    | −0.0139**<br>(0.0056)  | −0.0216***<br>(0.0059) | −0.0272***<br>(0.0080) | −0.0726<br>(0.0682)    | −0.0898**<br>(0.0386)  | −0.0507<br>(0.0337)    |
| Constant          | 0.0469<br>(0.0387)     | 0.0449***<br>(0.0167)  | 0.0916***<br>(0.0249)  | 0.0728**<br>(0.0309)   | 0.0435<br>(0.0371)     | −1.1010***<br>(0.1872) | −0.8888***<br>(0.1307) |
| R <sup>2</sup>    | 0.14                   | 0.11                   | 0.15                   | 0.097                  | 0.11                   | 0.14                   | 0.16                   |
| No. of firms      | 186                    | 186                    | 186                    | 186                    | 186                    | 186                    | 186                    |
| Observations      | 517                    | 517                    | 517                    | 517                    | 517                    | 516                    | 516                    |

Notes:

1. The numbers in parentheses are the heteroskedasticity-robust standard errors.

2. The asterisks \*\*\*, \*\*, and \* indicate 1%, 5%, and 10% of significance levels, respectively.

Our sample is drawn from three data sources. The data of waste and greenhouse gas emissions come from the Corporate Social Responsibility Database released by Toyo Keizai (2006, 2007, 2008, 2009). Since Toyo Keizai (2006) started to conduct questionnaire surveys on waste and greenhouse gas emissions, we use all available data in the initiation of this study. Furthermore, because the Toyo Keizai conducts multi-year questions in its survey, we utilize the results of the survey conducted at the most recent time.<sup>6</sup> Next, stock prices come from the “Kabuka” (Stock prices) CD-ROM 2010 provided by Toyo Keizai (2010). Finally, all other data are taken from the NEEDS (Nikkei Economic Electronic Databank System) released by Nikkei Digital Media (2010). Based on all available data of manufacturing firms from these three data sources, our sample consists of an unbalanced panel data, which includes 268 Japanese manufacturing firms from 2004 to 2008. Descriptive statistics of each variable are reported in Appendix Table A2.

As the estimation method, we employ the fixed effects model because our sample consists of five-year unbalanced panel data. Since the time horizon in our sample is relatively short, our study employs the industry-specific fixed effects estimation instead of the firm-specific fixed effects estimation.<sup>7</sup> The fixed effects estimation allows us to control unobserved industry-specific fixed effects that may affect financial performance and, therefore, deals with the endogeneity issues resulting from unobserved industry-specific effects.<sup>8</sup>

<sup>6</sup> Specifically, the data from 2004 comes from Toyo Keizai (2006); data from 2005 comes from Toyo Keizai (2007); data from 2006 comes from Toyo Keizai (2008); and data from 2007 and 2008 comes from Toyo Keizai (2009).

<sup>7</sup> Every firm in our sample in the manufacturing sector is classified into one of sixteen industries, listed in detail as shown in Appendix Table A3. This classification follows the sector sub-classification provided by the Securities Identification Code Committee (SICC). The SICC creates sector classification, based on the Japan Standard Industrial Classification defined by the Ministry of Internal Affairs and Communications, Japan.

<sup>8</sup> The firm-specific fixed effects estimation will capture the average level of dependent variables and the explanatory variables can only explain the deviations from five-year averages. On the contrary, the industry-specific fixed effects estimation will save the degree of freedom considerably, but can still capture the heterogeneity across industries.

Our analysis proceeds as follows. First, we report the estimation results using the full sample. Second, in capturing the relationship between the dirtiness of industries and environmental performance, we divide the full sample into ‘clean’ and ‘dirty’ industries since environmental performance is associated with the inherent dirtiness of an industry. We split the whole sample into nine clean and seven dirty industries, following Mani and Wheeler (1998), as shown in Appendix Table A3. For robustness checks, we distil each alternative dataset for the five cleanest and four dirtiest industries from the whole sample.<sup>9</sup> Finally, to assess the role of firm growth in the relationship between financial performance and environmental performance, we conduct the estimation, adding the interaction terms between environmental performance and firm growth.<sup>10,11</sup>

<sup>9</sup> Mani and Wheeler (1998) create the ranking of industrial sectors on actual emission intensity (emissions per unit of output) for conventional air pollutants, water pollutants, and heavy metals. Furthermore, since they also generate the overall ranking of industrial sectors, we employ this ranking (see Mani and Wheeler, 1998, Table 1). First of all, based on the top ten sectors with highest overall ranks, our all sample is divided into nine clean and seven dirty industries. “Oil and Coal Products” are classified as a clean industry, because Mani and Wheeler (1998) state “Petroleum has been excluded because a very few country are actually involved in its production”. Then, Mani and Wheeler (1998) select textiles, nonelectrical machinery, electrical machinery, transport equipment, and instruments as the five cleanest sectors, and iron and steel, nonferrous metals, industrial chemicals, pulp and paper, and metallic mineral products as the five dirtiest sectors. Thus, following this criterion, we create other datasets for the five cleanest and four dirtiest industries. Note that Mani and Wheeler (1998) employ the three-digit International Standard Industrial Classifications (ISIC), and this classification is not completely the same as ours.

<sup>10</sup> Konar and Cohen (2001) analyze the relationship between environmental performance on financial performance, taking into account the interaction terms between advertisement and firm growth, and between R&D and firm growth.

<sup>11</sup> Since our specification includes two environmental performance and some interaction terms at the same time, we have to care about the multicollinearity problems. The correlation coefficient between waste and greenhouse gas is 0.0780 and the variance inflation factors of every specification in all tables are less than ten. Therefore, we can conclude that our estimation does not have multicollinearity problems.

**Table 3**  
Environmental performance and financial performance (seven dirty industries).

|                   | (1)                 | (2)                    | (3)                    | (4)                    | (5)                   | (6)                    | (7)                    |
|-------------------|---------------------|------------------------|------------------------|------------------------|-----------------------|------------------------|------------------------|
|                   | ROE                 | ROA                    | ROI                    | ROIC                   | ROS                   | $q - 1$                | $\ln(q)$               |
| Size              | 0.0001<br>(0.0037)  | 0.0039<br>(0.0029)     | 0.0033<br>(0.0052)     | 0.0059<br>(0.0054)     | −0.0022<br>(0.0027)   | 0.2162***<br>(0.0374)  | 0.1407***<br>(0.0217)  |
| Growth            | 0.0002<br>(0.0003)  | 0.0002*<br>(0.0001)    | −0.0004**<br>(0.0002)  | −0.0005**<br>(0.0002)  | 0.0007<br>(0.0005)    | 0.0005<br>(0.0015)     | 0.0007<br>(0.0007)     |
| R&D               | −0.2134<br>(0.2178) | −0.0541<br>(0.0699)    | −0.0494<br>(0.1084)    | −0.0973<br>(0.1167)    | −0.1304<br>(0.3426)   | 1.1096<br>(1.4014)     | 0.7974<br>(0.7231)     |
| Advertisement     | −0.1127<br>(0.1280) | −0.0716<br>(0.0986)    | −0.2272<br>(0.1652)    | −0.1124<br>(0.1833)    | 0.1683<br>(0.1345)    | 1.5414<br>(1.6142)     | 1.7985**<br>(0.8854)   |
| Capital intensity | −0.0237<br>(0.0208) | 0.0023<br>(0.0049)     | 0.0098<br>(0.0067)     | 0.0157*<br>(0.0091)    | −0.0185*<br>(0.0097)  | 0.1032<br>(0.0685)     | 0.0556<br>(0.0442)     |
| Leverage          | 0.0000<br>(0.0089)  | −0.0196***<br>(0.0041) | −0.0412***<br>(0.0065) | −0.0460***<br>(0.0080) | −0.0160**<br>(0.0076) | −0.2504***<br>(0.0729) | −0.1271***<br>(0.0446) |
| Waste             | 0.0016<br>(0.0064)  | −0.0004<br>(0.0042)    | −0.0017<br>(0.0052)    | −0.0037<br>(0.0071)    | −0.0006<br>(0.0132)   | 0.0482<br>(0.0566)     | 0.0396<br>(0.0392)     |
| Greenhouse gas    | 0.1533<br>(0.2567)  | 0.0416<br>(0.1175)     | 0.0196<br>(0.1599)     | 0.0597<br>(0.2032)     | 3.6988***<br>(0.5902) | 1.6136<br>(1.9340)     | 1.1353<br>(1.1944)     |
| Constant          | 0.1024*<br>(0.0552) | 0.0566<br>(0.0362)     | 0.1214*<br>(0.0637)    | 0.1120*<br>(0.0654)    | 0.1323***<br>(0.0478) | −1.8751***<br>(0.3799) | −1.2849***<br>(0.2481) |
| R <sup>2</sup>    | 0.09                | 0.22                   | 0.29                   | 0.22                   | 0.67                  | 0.31                   | 0.36                   |
| No. of firms      | 82                  | 82                     | 82                     | 82                     | 82                    | 82                     | 82                     |
| Observations      | 234                 | 234                    | 234                    | 234                    | 234                   | 233                    | 233                    |

Notes:

1. The numbers in parentheses are the heteroskedasticity-robust standard errors.
2. The asterisks \*\*\*, \*\*, and \* indicate 1%, 5%, and 10% of significance levels, respectively.

### 3. Estimation Results

#### 3.1. Case of the Full Sample

Table 1 presents the estimation results using the full sample. The results imply that how environmental performance affects financial performance is obviously different for waste and greenhouse gas emissions. The effects of waste emission are not statistically significant on all financial performance in columns (1) through (7), whereas greenhouse gas emissions have significant negative impacts on ROA, ROI, ROIC, and Tobin's  $q - 1$  in columns (2), (3), (4), and (6). Since waste emissions are regulated by several laws such as the Waste Management and Public Cleansing Law in Japan, most firms have addressed waste emissions problems for a long period of time. Therefore, although there might have been some low hanging fruits in the past, because they have been exhaustively used so far, firms cannot acquire the additional benefit from waste reduction. In addition, because most firms usually operate by abiding to these regulations, it would be difficult to be superior to other firms in terms of further controlling waste emissions.<sup>12</sup> Also, the influence of waste emissions may be restricted to the relatively narrow range of stakeholders such as local residents. Thus, stakeholders such as stockholders, investors, and financial institutions cannot respond to increases and decreases in waste emissions as long as a firm follows the regulations.

Next, as for greenhouse gas emissions, their effects are significantly negative on ROA, ROI, and ROIC in columns (2) to (4). These results indicate that stakeholders such as investors, stockholders, and financial agencies take the long-run firm performance into account because ROA, ROI, and ROIC include the value of debt as well as equity capital. Specifically, due to the fact that greenhouse gas emissions may be regulated in the future, firms that seek to reduce them voluntarily in order to mitigate future regulation risks obtain higher evaluations from stakeholders. Likewise, in the sense that global warming issues

broadly affect human beings, they are related to a wider range of stakeholders unlike waste problems. Therefore, stakeholders may expect that firms addressing global warming issues will proactively improve their corporate reputation and image in the future. However, it is noteworthy that the effect of greenhouse gas emissions on ROE in column (1) is insignificant. This result may be attributed to the fact that ROE does not include debt but reflects equity capital. Also, although the coefficient of greenhouse gas on the natural logarithm of Tobin's  $q$  is not significant in column (7), its effect on Tobin's  $q - 1$  is significantly negative, suggesting that a decrease in greenhouse gas emissions leads to an increase in the value of intangible assets.

Furthermore, both waste and greenhouse gas emissions do not have a significant impact on ROS. Unlike other accounting-based financial performance, ROS does not include the capital structure in its calculation. Therefore, our results indicate that goods markets reflecting the evaluation by consumers and trading partners do not pay attention to a firms' efforts to reduce waste and greenhouse gas emissions. This result shows that stakeholders who do not have monetary relationships with firms seem to disregard environmental management as long as the firms operate by abiding to laws and regulations. While this result is not in line with that of Sarkis and Cordeiro (2001) who indicate negative impacts of pollution prevention and end-of-pipe efficiencies on ROS, it is similar to that of Wagner et al. (2002). Therefore, from Table 1, it is evident that the reactions of capital markets to the environmental performance of firms are different for each environmental issue.

As for other explanatory variables, the coefficients generally have expected signs. The effects of R&D on financial performance are significantly negative in some cases. This result can be attributed to the fact that our sample consists of five-year panel data and our estimations capture the short-run effects. Previous studies such as King and Lenox (2002) also report the negative relationship between R&D and financial performance.

#### 3.2. Cases of Clean and Dirty Industries

We investigate the effects of environmental performance on financial performance, dividing our sample of the manufacturing

<sup>12</sup> Hart and Ahuja (1996) state that although it is easier to remedy existing inefficiencies and waste emissions, once these processes are achieved, it is much more difficult to conduct further environmental management efforts.

**Table 4**  
Environmental performance and financial performance (five cleanest industries).

|                   | (1)                    | (2)                    | (3)                    | (4)                    | (5)                    | (6)                    | (7)                    |
|-------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                   | ROE                    | ROA                    | ROI                    | ROIC                   | ROS                    | $q - 1$                | $\ln(q)$               |
| Size              | 0.0070<br>(0.0045)     | 0.0022<br>(0.0016)     | 0.0011<br>(0.0022)     | 0.0027<br>(0.0029)     | 0.0032<br>(0.0072)     | 0.1297***<br>(0.0191)  | 0.0940***<br>(0.0121)  |
| Growth            | 0.2998***<br>(0.0541)  | 0.1072***<br>(0.0173)  | 0.1570***<br>(0.0247)  | 0.2006***<br>(0.0337)  | 0.2663**<br>(0.1129)   | 1.3791***<br>(0.2792)  | 0.9973***<br>(0.1897)  |
| R&D               | −0.2834<br>(0.1761)    | −0.0878<br>(0.0706)    | −0.1687<br>(0.1103)    | −0.1757<br>(0.1206)    | 0.1021<br>(0.9934)     | 0.8692<br>(1.0406)     | 0.4384<br>(0.6563)     |
| Advertisement     | −0.5966<br>(0.5765)    | 0.0621<br>(0.1851)     | 0.3256<br>(0.2687)     | 0.2139<br>(0.3787)     | −2.5697<br>(2.7010)    | 4.6418<br>(3.2743)     | 3.4879*<br>(2.1111)    |
| Capital intensity | −0.0482***<br>(0.0148) | −0.0006<br>(0.0025)    | −0.0023<br>(0.0033)    | 0.0073<br>(0.0058)     | −0.0320***<br>(0.0099) | −0.0455<br>(0.0345)    | −0.0195<br>(0.0218)    |
| Leverage          | 0.0100<br>(0.0143)     | −0.0109***<br>(0.0027) | −0.0170***<br>(0.0039) | −0.0208***<br>(0.0056) | 0.0109<br>(0.0099)     | −0.0415<br>(0.0329)    | −0.0170<br>(0.0206)    |
| Waste             | 0.0439<br>(0.0373)     | 0.0066<br>(0.0127)     | −0.0052<br>(0.0194)    | 0.0098<br>(0.0241)     | −0.0103<br>(0.2455)    | 0.0413<br>(0.2348)     | 0.0854<br>(0.1598)     |
| Greenhouse gas    | −0.0109<br>(0.0107)    | −0.0120***<br>(0.0033) | −0.0192***<br>(0.0046) | −0.0259***<br>(0.0057) | −0.0742<br>(0.0750)    | −0.1162**<br>(0.0505)  | −0.0880**<br>(0.0353)  |
| Constant          | 0.0445<br>(0.0417)     | 0.0576***<br>(0.0184)  | 0.1090***<br>(0.0278)  | 0.1021***<br>(0.0347)  | 0.0411<br>(0.0408)     | −1.0505***<br>(0.2003) | −0.8500***<br>(0.1375) |
| R <sup>2</sup>    | 0.30                   | 0.23                   | 0.26                   | 0.18                   | 0.21                   | 0.24                   | 0.28                   |
| No. of firms      | 136                    | 136                    | 136                    | 136                    | 136                    | 136                    | 136                    |
| Observations      | 366                    | 366                    | 366                    | 366                    | 366                    | 365                    | 365                    |

Notes:

1. The numbers in parentheses are the heteroskedasticity-robust standard errors.

2. The asterisks \*\*\*, \*\*, and \* indicate 1%, 5%, and 10% of significance levels, respectively.

industry into the two subsets of clean and dirty industries. This analysis allows us to examine the relationships between the dirtiness of an industry and its environmental performance. Following the classification criterion suggested by [Mani and Wheeler \(1998\)](#), every firm in our sample is first categorized into nine clean and seven dirty industries. Then, for robustness checks, we report the estimation results using the five cleanest and four dirtiest industries. The detailed classification and explanation of clean and dirty industries are described in [Footnote 9](#).

[Table 2](#) presents the results in the case of nine clean industries. The sign and significance of the coefficients of waste and greenhouse gas emissions are almost similar to those in the case of the full sample in [Table 1](#). That is to say, waste emissions do not have significant impacts on financial performance, whereas greenhouse gas emissions have significantly negative effects on financial performance. This is partly because the number of firms which are classified into clean industries in our sample is larger than that of dirty industries. One difference between the full sample and nine industries is that the effect of waste

**Table 5**  
Environmental performance and financial performance (four dirtiest industries).

|                   | (1)                   | (2)                    | (3)                    | (4)                    | (5)                    | (6)                    | (7)                    |
|-------------------|-----------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                   | ROE                   | ROA                    | ROI                    | ROIC                   | ROS                    | $q - 1$                | $\ln(q)$               |
| Size              | −0.0054<br>(0.0048)   | −0.0011<br>(0.0032)    | −0.0056<br>(0.0056)    | −0.0024<br>(0.0059)    | −0.0081***<br>(0.0030) | 0.1899***<br>(0.0429)  | 0.1203***<br>(0.0243)  |
| Growth            | 0.1176***<br>(0.0425) | 0.0567***<br>(0.0197)  | 0.0797**<br>(0.0311)   | 0.0732**<br>(0.0346)   | 0.1323**<br>(0.0520)   | 0.3032<br>(0.3097)     | 0.2768<br>(0.2098)     |
| R&D               | −0.1334<br>(0.2956)   | −0.1583<br>(0.1754)    | −0.3181<br>(0.2755)    | −0.4401<br>(0.3797)    | 0.3610*<br>(0.2163)    | −2.5399<br>(2.7215)    | −0.3266<br>(1.6527)    |
| Advertisement     | 0.1539<br>(0.1384)    | 0.1550<br>(0.1084)     | 0.1719<br>(0.1684)     | 0.3221<br>(0.2229)     | 0.3179***<br>(0.1100)  | 4.6197***<br>(1.5701)  | 3.7929***<br>(0.8343)  |
| Capital intensity | −0.0304<br>(0.0268)   | −0.0047<br>(0.0059)    | −0.0042<br>(0.0082)    | 0.0025<br>(0.0111)     | −0.0180<br>(0.0118)    | 0.0214<br>(0.0848)     | 0.0133<br>(0.0559)     |
| Leverage          | 0.0078<br>(0.0127)    | −0.0129***<br>(0.0045) | −0.0284***<br>(0.0068) | −0.0350***<br>(0.0092) | −0.0089<br>(0.0109)    | −0.2166**<br>(0.0835)  | −0.0962*<br>(0.0517)   |
| Waste             | 0.0027<br>(0.0068)    | 0.0004<br>(0.0044)     | 0.0003<br>(0.0056)     | −0.0019<br>(0.0075)    | 0.0002<br>(0.0106)     | 0.0554<br>(0.0570)     | 0.0448<br>(0.0392)     |
| Greenhouse gas    | 0.1200<br>(0.3034)    | 0.0908<br>(0.1821)     | 0.1326<br>(0.2559)     | 0.2382<br>(0.3546)     | 3.4078***<br>(0.4824)  | 3.9905<br>(2.4402)     | 1.7502<br>(1.5017)     |
| Constant          | 0.1465**<br>(0.0714)  | 0.1076**<br>(0.0423)   | 0.2154***<br>(0.0746)  | 0.2067***<br>(0.0765)  | 0.1504***<br>(0.0405)  | −1.4006***<br>(0.4361) | −1.0228***<br>(0.2900) |
| R <sup>2</sup>    | 0.13                  | 0.18                   | 0.24                   | 0.19                   | 0.85                   | 0.29                   | 0.34                   |
| No. of firms      | 59                    | 59                     | 59                     | 59                     | 59                     | 59                     | 59                     |
| Observations      | 159                   | 159                    | 159                    | 159                    | 159                    | 158                    | 158                    |

Notes:

1. The numbers in parentheses are the heteroskedasticity-robust standard errors.

2. The asterisks \*\*\*, \*\*, and \* indicate 1%, 5%, and 10% of significance levels, respectively.

**Table 6**  
Environmental performance and financial performance with interaction terms (full sample).

|                         | (1)                    | (2)                    | (3)                    | (4)                    | (5)                    | (6)                    | (7)                    |
|-------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                         | ROE                    | ROA                    | ROI                    | ROIC                   | ROS                    | $q-1$                  | $\ln(q)$               |
| Size                    | 0.0045<br>(0.0029)     | 0.0030**<br>(0.0012)   | 0.0020<br>(0.0019)     | 0.0046**<br>(0.0023)   | 0.0011<br>(0.0040)     | 0.1504***<br>(0.0154)  | 0.1061***<br>(0.0097)  |
| Growth                  | 0.0003<br>(0.0003)     | 0.0003***<br>(0.0001)  | −0.0002<br>(0.0002)    | −0.0001<br>(0.0003)    | 0.0016<br>(0.0021)     | 0.0035**<br>(0.0017)   | 0.0027**<br>(0.0011)   |
| R&D                     | −0.2959***<br>(0.1073) | −0.1027**<br>(0.0411)  | −0.1561**<br>(0.0654)  | −0.2047***<br>(0.0722) | 0.1151<br>(0.4761)     | 0.7664<br>(0.6668)     | 0.4792<br>(0.4193)     |
| Advertisement           | −0.2174<br>(0.1767)    | 0.0741<br>(0.0777)     | 0.0221<br>(0.1077)     | 0.2077<br>(0.1472)     | −0.1381<br>(0.5401)    | 3.2125***<br>(1.0589)  | 2.5884***<br>(0.6191)  |
| Capital intensity       | −0.0317***<br>(0.0103) | 0.0012<br>(0.0021)     | 0.0034<br>(0.0029)     | 0.0120**<br>(0.0051)   | −0.0301***<br>(0.0064) | −0.0162<br>(0.0273)    | −0.0152<br>(0.0149)    |
| Leverage                | 0.0026<br>(0.0115)     | −0.0127***<br>(0.0023) | −0.0238***<br>(0.0035) | −0.0277***<br>(0.0051) | 0.0136**<br>(0.0062)   | −0.0830***<br>(0.0319) | −0.0282<br>(0.0172)    |
| Waste                   | −0.0032<br>(0.0024)    | 0.0011<br>(0.0011)     | −0.0002<br>(0.0018)    | −0.0002<br>(0.0022)    | 0.0304<br>(0.0224)     | 0.0191<br>(0.0173)     | 0.0212<br>(0.0131)     |
| Waste × Growth          | −0.0026<br>(0.0028)    | −0.0024**<br>(0.0008)  | 0.0002<br>(0.0013)     | −0.0018<br>(0.0015)    | −0.0366<br>(0.0232)    | −0.0416***<br>(0.0130) | −0.0274***<br>(0.0083) |
| Greenhouse gas          | −0.0165<br>(0.0200)    | −0.0160***<br>(0.0037) | −0.0242***<br>(0.0053) | −0.0303***<br>(0.0065) | −0.0789<br>(0.1136)    | −0.1124<br>(0.0751)    | −0.0598<br>(0.0586)    |
| Greenhouse gas × Growth | 0.1029<br>(0.1909)     | 0.1191***<br>(0.0428)  | 0.0840*<br>(0.0490)    | 0.1302*<br>(0.0719)    | 3.1126<br>(2.7875)     | 1.8501*<br>(0.9966)    | 1.0367<br>(0.6810)     |
| Constant                | 0.0631*<br>(0.0324)    | 0.0489***<br>(0.0150)  | 0.1022***<br>(0.0242)  | 0.0848***<br>(0.0277)  | 0.0470*<br>(0.0264)    | −1.2630***<br>(0.1633) | −0.9712***<br>(0.1125) |
| R <sup>2</sup>          | 0.13                   | 0.14                   | 0.20                   | 0.12                   | 0.19                   | 0.18                   | 0.21                   |
| No. of firms            | 268                    | 268                    | 268                    | 268                    | 268                    | 268                    | 268                    |
| Observations            | 751                    | 751                    | 751                    | 751                    | 751                    | 749                    | 749                    |

Notes:

1. The numbers in parentheses are the heteroskedasticity-robust standard errors.
2. The asterisks \*\*\*, \*\*, and \* indicate 1%, 5%, and 10% of significance levels, respectively.

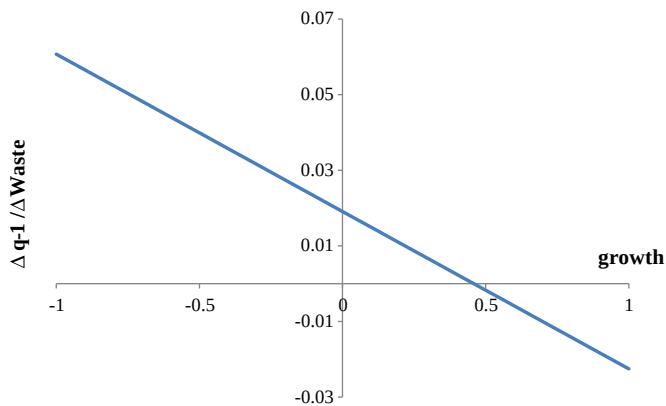
emissions on ROE is significantly negative in clean industries in column (1).

Table 3 illustrates the results when our sample is reduced to the firms in seven dirty industries. In this case, both waste and greenhouse gas emissions do not have significant impacts on financial performance. This outcome may result from the fact that waste problems are considered to be more important in dirty industries than global warming issues among various stakeholders. In addition to the reasons for the insignificance of waste emissions in the full sample as mentioned in the previous subsection, the damage resulting from waste problems seems to be larger in dirty industries than in clean industries. Since a dirty industry firm not

only suffers from legal penalties but loses its reputation if it fails to abide by the regulations, it recognizes these risks and complies with the regulations. Thus, waste emissions do not have significant effects on financial performance, as far as a firm follows the regulations. On the contrary, since reductions in greenhouse gas emissions are not necessarily obligated by the present laws, reductions depend on the firm's voluntary approach. Therefore, various stakeholders still do not pay attention to reductions in greenhouse gas emissions in dirty industries at this point.<sup>13</sup>

For robustness checks, we also report the estimation results using the five cleanest and four dirtiest industries in Tables 4 and 5, respectively. The results are almost similar to those in Tables 2 and 3, and therefore, our estimation is robust. Waste emissions do not have significant impacts on financial performance in both clean and dirty industries. On the other hand, a decrease in greenhouse gas emissions improves financial performance in clean industries, whereas greenhouse gas does not influence financial performance in dirty industries. These results may imply that problems of greenhouse gas emissions are different from those of waste laying which is regulated strictly by laws and regulations.

In sum, waste emissions do not have significant impacts on most financial performance in the case of the full sample, clean industries, and dirty industries, suggesting that various stakeholders do not necessarily value environmental management of waste emissions positively and negatively. On the other hand, while the effects of greenhouse gas emissions are generally significantly negative on financial performance in the whole sample and clean industries, their effects are not statistically

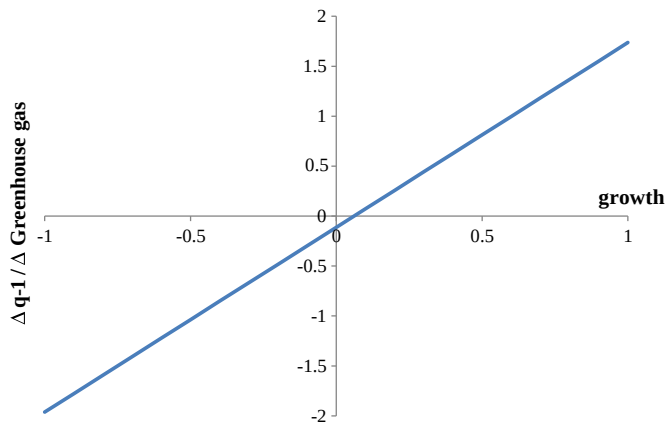


Notes:

1. This figure is illustrated based on the result of column (6), Table 6.

**Fig. 1.** The partial effect of waste on Tobin's  $q-1$ .

<sup>13</sup> This result would hold even in the goods markets, because the effect of greenhouse gas emissions on ROS is significant in column (5) due to some outliers. To be precise, when we exclude five outliers, the greenhouse gas coefficient is not significant.



Notes:

1. This figure is illustrated based on the result of column (6), Table 6.

Fig. 2. The partial effect of greenhouse gas on Tobin's  $q - 1$ .

significant in dirty industries. The estimation results provide mixed relationships between environmental and financial performance, implying that the evaluation of each environmental performance is different among various financial performance. This result can be accounted for by the responses of stakeholders having different interests in varying markets.

### 3.3. Interaction between Environmental Performance and Firm Growth

To capture the role of firm growth in the relationship between environmental performance and financial performance, we add two interaction terms between firm growth and environmental performance, waste and greenhouse gas. Table 6 reports the results in the case of the full sample with these two interaction terms. In columns (2), (6), and (7) where the dependent variables are ROA, Tobin's  $q - 1$ , and the natural logarithm of Tobin's  $q$ , the interaction terms between waste and firm growth have significantly negative impacts. Although the coefficient of waste is not significant, the null hypothesis that both coefficients of waste and its interaction term with firm growth are simultaneously zero is rejected at 1% significance level in the F test in columns (2), (6), and (7). Thus, the partial effects of waste emissions on financial performance decrease as firm growth increases. As discussed above, waste emissions do not have significant impact on financial performance without interaction terms. However, considering the interaction terms between waste emissions and firm growth, the partial effects of waste emissions are meaningful.

In contrast, in columns (2), (3), (4), and (6), where the dependent variables are ROA, ROI, ROIC, and Tobin's  $q - 1$ , the interaction terms between greenhouse gas emissions and firm growth have significant positive impacts on financial performance. Although the coefficient of greenhouse gas emissions is not significant, we can reject the null hypothesis that both coefficients of greenhouse gas and its interaction term with firm growth are simultaneously zero at the conventional significance level in the F test in columns (2), (3), (4), and (6).

In order to comprehensively demonstrate these results in more detail, we illustrate the figures regarding the partial effects of waste and greenhouse gas on Tobin's  $q - 1$  in column (6) in Figs. 1 and 2. As firm growth increases, the partial effect of waste decreases, whereas the partial effect of greenhouse gas increases. The threshold level of firm growth separating the negative and positive effects of environmental performance on Tobin's  $q - 1$  is 0.4588 and 0.0608 in the case of waste and

greenhouse gas, respectively.<sup>14</sup> In the case of waste emissions in Fig. 1, its partial effect is positive in most firms because the number of firms whose growth rate is above the threshold value is only 7 firms. This outcome may result from the fact that since waste emissions are regulated by laws, most environmental regulation-abiding firms may have difficulty conducting further environmental management efforts. However, this positive effect is mitigated as firm growth rate increases. In contrast, in the case of greenhouse gas emissions in Fig. 2, its partial effect is positive if firm growth is positive. One possible explanation is that since greenhouse gas emissions are not explicitly regulated, the markets highly value firms that prioritize an increase in production over a reduction in greenhouse gas emissions when these firms are growing rapidly.

## 4. Conclusion

This paper examines the relationship between environmental and financial performance using the data of Japanese manufacturing firms from 2004 to 2008. Since environmental issues have been diverse in recent years, each stakeholder may have different preferences for each environmental issue. Given this, this study employs seven financial performance indices reflecting each market evaluation in order to clarify how each market evaluates corporate management when dealing with different environmental issues. Furthermore, in order to test the hypothesis that each stakeholder may emphasize different environmental issues, we utilize the amount of waste and greenhouse gas emissions as the proxies for environmental performance. To the best of our knowledge, this paper is the first in attempting to use emissions of waste and greenhouse gas instead of environmental management scores from studies using Japanese firm-level data.

Our estimation results show that the responses of financial performance are different depending on each environmental issue, the results of which are attributed to varying stakeholder preferences. For example, the effects of waste emissions on financial performance are generally not statistically significant. On the other hand, greenhouse gas reduction increases most long-run financial performance, whereas it does not have a significant effect on ROS which shows the short-run financial performance. In other words, stakeholders such as investors, stockholders, and financial agencies take the long-run firm performance into account, but stakeholders such as consumers and trading partners who are closely related to the goods market do not care about corporate environmental management in the short-run.

Although our results are new to the existing literature, there is a caveat to our sample. Our sample consists of five-year unbalanced panel data and the time horizon is relatively short, although we employ all available data in the initiation of this study. Therefore, our sample may not be enough to identify the long-run relationship between environmental and financial performance. An investigation using a longer panel data may be a promising future research study.

Our results have some policy implications for the relationship between firms and the government. Although environmental problems have mainly been left to government intervention to solve them, if firms have incentives to reduce their environmental damages, the market mechanism could provide some clues to deal with environmental issues. Furthermore, when the market mechanism alone cannot solve environmental issues, the government should formulate appropriate regulations and laws to complement it.

## Acknowledgement

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<sup>14</sup> In this analysis, the number of observations such that firm growth is below the threshold is 742 and 502 out of 749 observations in the case of waste and greenhouse gas, respectively.

## Appendix. Data Definitions and Sources

**Table A1**

Data definitions and sources.

| Variable          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Source                                                               |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| ROE               | Return on equity is defined as income for the term divided by total stockholders' equity (average of 2 terms).                                                                                                                                                                                                                                                                                                                                                                        | Nikkei Digital Media (2010)                                          |
| ROA               | Return on assets is defined by the following ratio. The numerator is the sum of operating profit, interest revenue/discount fee/interest on securities, and dividend revenue. The denominator is the sum of liabilities and net assets (average of 2 terms).                                                                                                                                                                                                                          | Nikkei Digital Media (2010)                                          |
| ROI               | Return on investment is defined by the following ratio. The numerator is the same as that in the case of ROA. The denominator is [total assets – (total current assets – total current liabilities)].                                                                                                                                                                                                                                                                                 | Nikkei Digital Media (2010)                                          |
| ROIC              | Return on invested capital is defined by the following ratio. The numerator is the same as that in the case of ROA. The denominator is {total stockholders' equity + short-term borrowings + CPS + long-term borrowings within 1 year + corporate bonds due within 1 year and CBS + deposits from employees + corporate bonds and CBS + long-term borrowings + discounted notes receivable – (cash and equivalents + securities + business-account loans and investment securities)}. | Nikkei Digital Media (2010)                                          |
| ROS               | Return on sales is defined as income for the term divided by sales and operating revenue.                                                                                                                                                                                                                                                                                                                                                                                             | Nikkei Digital Media (2010)                                          |
| $q - 1$           | This variable is Tobin's $q - 1$ . Tobin's $q$ is defined as (total debt + stock prices $\times$ the number of stocks)/total assets.                                                                                                                                                                                                                                                                                                                                                  | Nikkei Digital Media (2010) and Toyo Keizai (2010)                   |
| $\ln(q)$          | This variable is the natural logarithm of Tobin's $q$ .                                                                                                                                                                                                                                                                                                                                                                                                                               | Nikkei Digital Media (2010) and Toyo Keizai (2010)                   |
| Size              | Firm size is the natural logarithm of sales and operating revenue (in the unit of million Japanese Yen).                                                                                                                                                                                                                                                                                                                                                                              | Nikkei Digital Media (2010)                                          |
| Growth            | Firm growth is the growth rate in sales and operating revenue.                                                                                                                                                                                                                                                                                                                                                                                                                        | Nikkei Digital Media (2010)                                          |
| R&D               | Research and development intensity is defined as research and development expenses divided by sales and operating revenue.                                                                                                                                                                                                                                                                                                                                                            | Nikkei Digital Media (2010)                                          |
| Advertisement     | Advertisement intensity is defined as advertisement expenses divided by sales and operating revenue.                                                                                                                                                                                                                                                                                                                                                                                  | Nikkei Digital Media (2010)                                          |
| Capital intensity | Capital intensity is defined as sales and operating revenue divided by stockholders' equity.                                                                                                                                                                                                                                                                                                                                                                                          | Nikkei Digital Media (2010)                                          |
| Leverage          | Financial leverage is defined as the sum of liabilities and net assets (average of 2 terms) is divided by total stockholders' equity.                                                                                                                                                                                                                                                                                                                                                 | Nikkei Digital Media (2010)                                          |
| Waste             | This variable is defined as waste emissions (in the unit of tons) divided by sales and operating revenue (in the unit of million Japanese Yen).                                                                                                                                                                                                                                                                                                                                       | Nikkei Digital Media (2010) and Toyo Keizai (2006, 2007, 2008, 2009) |
| Greenhouse gas    | This variable is defined as greenhouse gas emissions (in the unit of 1,000 tons of CO <sub>2</sub> equivalents) divided by sales and operating revenue (in the unit of million Japanese Yen).                                                                                                                                                                                                                                                                                         | Nikkei Digital Media (2010) and Toyo Keizai (2006, 2007, 2008, 2009) |

**Table A2**

Descriptive statistics.

| Variables         | Mean    | Std. Dev. | Min.    | Max.    |
|-------------------|---------|-----------|---------|---------|
| ROE               | 0.0423  | 0.1001    | −0.9432 | 0.3069  |
| ROA               | 0.0557  | 0.0415    | −0.0761 | 0.2133  |
| ROI               | 0.0740  | 0.0658    | −0.0915 | 0.4349  |
| ROIC              | 0.0969  | 0.0828    | −0.1106 | 0.6952  |
| ROS               | 0.0377  | 0.1381    | −1.7789 | 0.7633  |
| $q - 1$           | 0.4021  | 0.6021    | −0.6350 | 4.6476  |
| $\ln(q)$          | 0.2642  | 0.3713    | −1.0078 | 1.7312  |
| Size              | 11.8980 | 1.4937    | 7.9288  | 16.3070 |
| Growth            | 0.2445  | 4.4049    | −0.9297 | 91.9909 |
| R&D               | 0.0498  | 0.0546    | 0.0001  | 0.4047  |
| Advertisement     | 0.0134  | 0.0203    | 0.0000  | 0.1348  |
| Capital intensity | 1.9535  | 1.3862    | 0.0570  | 10.5098 |
| Leverage          | 2.1291  | 1.0471    | 1.0700  | 11.3700 |
| Waste             | 0.1717  | 0.6809    | 0       | 13.7018 |
| Greenhouse gas    | 0.0084  | 0.0924    | 0       | 2.1909  |

**Table A3**

Detailed classification in manufacturing sector.

| Clean                        | Dirty                 |
|------------------------------|-----------------------|
| Foods                        | Pulp and Paper (D)    |
| Textiles and Apparels (C)    | Chemicals (D)         |
| Oil and Coal Products        | Pharmaceutical        |
| Glass and Ceramics Products  | Rubber Products       |
| Machinery (C)                | Iron and Steel (D)    |
| Electric Appliances (C)      | Nonferrous Metals (D) |
| Transportation Equipment (C) | Metal Products        |
| Precision Instruments (C)    |                       |
| Other Products               |                       |

Notes:

1. "C" and "D" in parentheses indicate industries which include the five cleanest and four dirtiest industries, respectively.

## References

- Ambec, S., Lanoie, P., 2008. Does it pay to be green? a systematic overview. *Academy of Management Perspectives* 22 (4), 45–62.
- Blanco, E., Rey-Maqueieira, J., Lozano, J., 2009. The economic impacts of voluntary environmental performance of firms: a critical review. *Journal of Economic Surveys* 23 (3), 462–502.
- Butz, C., Plattner, A., 2000. *Socially Responsible Investment: A Statistical Analysis of Returns*. Bank Sarasin Basic Report, Basel.
- Cole, M.A., Elliott, R.J.R., Shimamoto, K., 2006. Globalization, firm-level characteristics and environmental management: a study of Japan. *Ecological Economics* 59 (3), 312–323.
- Cordeiro, J.J., Sarkis, J., 1997. Environmental proactivism and firm performance: evidence from security analyst earnings forecasts. *Business Strategy and the Environment* 6 (2), 104–114.
- Elsayed, K., Paton, D., 2005. The impact of environmental performance on firm performance: static and dynamic panel data evidence. *Structural Change and Economic Dynamics* 16 (3), 395–412.
- Hart, S.L., 1995. A natural-resource-based view of the firm. *Academy of Management Review* 20 (4), 986–1014.
- Hart, S.L., Ahuja, G., 1996. Does it pay to be green? an empirical examination of the relationship between emission reduction and firm performance. *Business Strategy and the Environment* 5 (1), 30–37.
- Hirsch, B.T., Seaks, T.G., 1993. Functional form in regression models of Tobin's  $q$ . *Review of Economics and Statistics* 75 (2), 381–385.
- King, A., Lenox, M., 2002. Exploring the locus of profitable pollution reduction. *Management Science* 48 (2), 289–299.
- Konar, S., Cohen, M.A., 2001. Does the market value environmental performance? *Review of Economics and Statistics* 83 (2), 281–289.
- Mani, M., Wheeler, D., 1998. In search of pollution havens? dirty industry in the world economy, 1960 to 95. *Journal of Environment and Development* 7 (3), 215–247.
- Molina-Azorín, J.F., Claver-Cortés, E., López-Gamero, M.D., Tarí, J.J., 2009. Green management and financial performance: a literature review. *Management Decision* 47 (7), 1080–1100.
- Murphy, C.J., 2002. *The Profitable Correlation between Environmental and Financial Performance: A Review of the Research*. Light Green Advisors, Seattle.
- Nagayama, S., Takeda, F., 2007. An empirical study on the impact of environmentally-friendly news on stock prices in Japan. *Global Business and Economics Anthology* 1, 129–137.
- Nakao, Y., Amano, A., Matsumura, K., Genba, K., Nakano, M., 2007. Relationship between environmental performance and financial performance: an empirical analysis of Japanese corporations. *Business Strategy and the Environment* 16 (2), 106–118.

- Nikkei Digital Media, 2010. NEEDS Financial Quest 2.0. Nikkei Digital Media, Tokyo. (online).
- Porter, M.E., van der Linde, C., 1995. Toward a new conception of the environment-competitiveness relationship. *Journal of Economic Perspectives* 9 (4), 97–118.
- Rassier, D.G., Earnhart, D., 2010. Does the Porter hypothesis explain expected future financial performance? the effect of clean water regulation on chemical manufacturing firms. *Environmental and Resource Economics* 45 (3), 353–377.
- Russo, M.V., Fouts, P.A., 1997. A resource-based perspective on corporate environmental performance and profitability. *Academy of Management Journal* 40 (3), 534–559.
- Salama, A., 2005. A note on the impact of environmental performance on financial performance. *Structural Change and Economic Dynamics* 16 (3), 413–421.
- Sarkis, J., Cordeiro, J.J., 2001. An empirical evaluation of environmental efficiencies and firm performance: pollution prevention versus end-of-pipe practice. *European Journal of Operational Research* 135 (1), 102–113.
- Takeda, F., Tomozawa, T., 2008. A change in market responses to the environmental management ranking in Japan. *Ecological Economics* 67 (3), 465–472.
- Toyo Keizai, 2006. Corporate Social Responsibility Database 2007. Toyo Keizai, Tokyo. (in Japanese).
- Toyo Keizai, 2007. Corporate Social Responsibility Database 2008. Toyo Keizai, Tokyo. (in Japanese).
- Toyo Keizai, 2008. Corporate Social Responsibility Database 2009. Toyo Keizai, Tokyo. (in Japanese).
- Toyo Keizai, 2009. Corporate Social Responsibility Database 2010. Toyo Keizai, Tokyo. (in Japanese).
- Toyo Keizai, 2010. Kabuka (Stock prices) CD-ROM 2010. Toyo Keizai, Tokyo. (in Japanese).
- Wagner, M., Van Phu, N., Azomahou, T., Wehrmeyer, W., 2002. The relationship between the environmental and economic performance of firms: an empirical analysis of the European paper industry. *Corporate Social Responsibility and Environmental Management* 9 (3), 133–146.
- Walley, N., Whitehead, B., 1994. It's not easy being green. *Harvard Business Review* 72 (3), 46–51.
- Yamaguchi, K., 2008. Reexamination of stock price reaction to environmental performance: a GARCH application. *Ecological Economics* 68 (1–2), 345–352.